

USAMA BIN NADEEM

Software Engineer | Python Backend | FastAPI · Django · REST APIs | AI & Automation
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PROFESSIONAL SUMMARY

Software engineer with 2+ years of professional Python experience across freelance, product and software-agency environments, specialising in backend services and REST APIs built with FastAPI, Django REST Framework and Flask. Delivers LLM-powered chatbot backends, inventory and data-management systems, and Selenium/BeautifulSoup automation pipelines backed by PostgreSQL and MySQL. Owns features end to end — API design, database schema, third-party integration and client hand-off. Currently based in Malaysia as an international postgraduate student; open to software engineering roles in Kuala Lumpur, subject to applicable work authorisation.

CORE TECHNICAL SKILLS

Languages: Python, SQL

Backend & APIs: FastAPI, Django, Django REST Framework, Flask, REST API design, authentication & user management, third-party API integration

Databases: PostgreSQL, MySQL, relational schema design, SQL queries, data modelling

AI & Machine Learning: LLM integration, conversational AI / chatbot development, machine learning & deep learning (age & gender detection from images)

Automation & Data: Selenium, BeautifulSoup, web scraping, data extraction & processing, SMTP / email automation

Data Visualisation: Streamlit, Apache Superset dashboard integration

Tools & Platforms: Git, GitHub; Shopify app development and WordPress customisation (secondary)

PROFESSIONAL EXPERIENCE

Freelance Software Engineer — Backend & AI — Remote | April 2025 – Present

- Build FastAPI backend services for production chatbot applications, exposing REST endpoints that clients integrate into their own web and messaging products.
- Design and integrate conversational AI systems powered by Large Language Models (LLMs), including an AI chatbot delivered across the Katrix and Suna platforms.
- Develop RESTful APIs with Django REST Framework for multiple client projects — endpoint design, serialisation, authentication and database integration.
- Integrated Apache Superset dashboards into a Django REST project so client teams could explore operational data directly instead of requesting ad-hoc reports.
- Work directly with clients to translate business requirements into scoped technical solutions, and support the delivered services after hand-off.

Backend Developer — Blagle Impex — Sialkot, Pakistan | June 2024 – March 2025

- Developed Python/Flask backend applications, including an inventory management system with MySQL integration for tracking products and stock records.
- Built an Amazon product web-scraping service that extracted listing, attribute and pricing data and fed it into the company's inventory workflow.
- Developed a custom product builder as a Shopify app, extending store functionality beyond stock theme capability.
- Worked extensively with relational databases to store, retrieve and manipulate business data, writing queries and handling data integrity.
- Maintained and customised Shopify and WordPress storefronts, integrating them with backend systems alongside the team.

Associate Software Engineer — InvoZone | February 2024 – May 2024

- Developed and maintained Django-based applications, implementing new features and structuring code for maintainability as the systems grew.
- Built interactive Streamlit data-visualisation tools that made datasets explorable for non-technical stakeholders.
- Reviewed teammates' code and worked with cross-functional teams to deliver releases.

Software Engineering Intern — Web Scraping & Automation — InvoZone | November 2023 – January 2024

- Built Python web-scraping and data-extraction scripts using BeautifulSoup and Selenium.
- Integrated scraped datasets into existing systems and debugged extraction failures caused by changes in site structure.

SELECTED PROJECTS

Full-Stack Social Media Platform *Python · FastAPI · MongoDB · Azure Blob Storage*

- Built a Twitter-style social platform on FastAPI and MongoDB, with posts, retweets, following, profile editing and search.
- Handled media storage through Azure Blob Storage and designed the document schema around the platform's read-heavy access patterns.

LSTM vs GRU Next-Word Prediction for Assistive Communication *Python · Deep Learning · Streamlit*

- Trained a next-word prediction system on the DailyDialog dataset to support assistive communication use cases.
- Compared LSTM and GRU architectures head-to-head and reported real measured results rather than estimates, with a Streamlit interface for interactive testing.

JobPilot — AI-Assisted Job Search and Application Tracking Platform *Full-stack · AI-assisted build*

- Specified and directed a full-stack platform for finding compliant job sources, tailoring a CV per job, and tracking every application in one place.
- Chose a compliance-first design over naive scraping, defining the data sources and architecture rather than hand-writing the implementation.

Student Record Management System with Role-Based Access *Python · Streamlit · SQLite*

- Built a role-based student management system with separate Admin, Teacher and Student permissions over shared records.
- Implemented student CRUD, an analytics dashboard and bulk Excel import for onboarding existing cohorts.

Multi-Source Sales Lead Automation Pipeline *Python · Web Scraping · SMTP validation*

- Built a production lead-generation pipeline at InvoZone that scraped job postings across multiple portals.
- Extracted decision-makers, validated contacts and discovered verified email addresses automatically, replacing manual prospecting.

Real-Time Gender Detection System *Python · OpenCV · Deep Learning · Computer Vision*

- Classified gender from a live webcam feed in real time, using OpenCV for face detection and a trained convolutional neural network for classification.

LLM-Powered Chatbot Platform *Python · FastAPI · REST APIs · LLM integration*

- Clients needed conversational assistants embedded inside their existing products rather than a separate chat tool.
- Built the FastAPI backend and REST endpoints handling conversation requests, session flow and LLM provider calls.
- Delivered the assistant as an integrated chatbot service on the Katrix and Suna platforms.

AI-Enabled Smart Advertisement System (Final Year Project) *Python · Machine Learning · Deep Learning · Computer Vision*

- Digital advertising displays typically show identical content to every viewer, regardless of who is watching.
- Built a system that detects a viewer's age and gender from camera input using deep learning models and selects the advertisement shown accordingly.

Inventory Management System & REST API *Python · Flask · MySQL · Django REST Framework*

- Built a Flask and MySQL inventory system for tracking products, stock levels and transaction records.
- Re-implemented the same domain as a RESTful service with Django REST Framework, exposing CRUD endpoints over a relational schema.

Amazon Product Data Extraction Pipeline *Python · Flask · Selenium · BeautifulSoup · MySQL*

- Built a Flask application that extracts Amazon product data — listings, attributes and pricing — on demand.
- Handled dynamically rendered pages with Selenium and structured HTML parsing with BeautifulSoup, persisting results to a database for downstream inventory use.

Authentication & Email Verification Services *Python · FastAPI · Django · Flask · SMTP*

- Implemented a FastAPI authentication service covering user registration, secure login and user management.
- Built email validation and verification services (Django, and Flask over SMTP) to confirm address deliverability and protect data integrity at sign-up.

EDUCATION

Master of Computer Science (Information Security and Assurance) — *Universiti Sains Islam Malaysia* | Master by Mixed Mode — Sep 2026 – Mar 2028

Bachelor of Science in Computer Engineering — *University of Management and Technology* — *Sialkot, Pakistan* | 2019 – 2023

- Final Year Project: AI-Enabled Smart Advertisement System (machine learning / deep learning).

ADDITIONAL INFORMATION

Languages: English, Urdu

Location: Currently based in Malaysia | International student | Open to Kuala Lumpur software engineering opportunities subject to applicable work authorisation.